

Geometry supervision for 3DGRUT

Depth and normal quality on OB3D: measurements

7k iterations, OB3D, three scenes

Six candidate supervision terms, each reported as

PREMISE	EXPERIMENT	RESULT	QUALITATIVE	CONCLUSION
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SCENES OB3D, with reference depth and normals:

sponza, lone-monk, emerald-square.

PROTOCOL 7k iterations. Reference geometry is read by evaluation only; no training term uses it. Held-out split for all metrics.

SEED NOISE over `seed_initialization` 1–4:

- PSNR ± 0.04 – 0.13 dB
- `abs_rel` ± 0.002 – 0.004
- normal ± 0.1 – 1.2°

Single-term cells are 3-seed; combination cells are single-seed.

METRICS

- `abs_rel` — mean $|d - d^*|/d^*$ over pixels where the reference has depth and the ray is opaque enough to have one.
- `normalo` — mean angle to the reference normal, signed (a flipped normal scores 180° , not 0°).
- `n_gain` — **control**: `normalo` of the view-direction control minus `normalo` of the render. The control points every normal back along the view ray and uses no geometry.
- PSNR — on the composited image, as validation reports it.

`n_gain` < 0: the normal buffer is worse than using no geometry at all.

Reference runs: both primitives, no geometry term

Run	sponza				lone-monk				emerald-square			
	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR
gaussian (ref)	0.0571	52.4	-9.6	36.29	0.0982	54.3	-19.1	36.72	0.1368	65.8	-15.5	32.90
trisurfel (ref)	0.0600	44.7	-1.9	36.19	0.0987	35.0	+0.2	36.71	0.1327	49.4	+0.9	32.92

RESULT

- PSNR is within 0.03 dB between primitives on every scene.
- gaussian has the better abs_rel on sponza and lone-monk; trisurfel on emerald-square.
- trisurfel has the better normals on **all three**, by 7.7–19.3^o.
- gaussian's n_gain is -9.6 to -19.1. trisurfel's is -1.9 to +0.9.

CONCLUSION

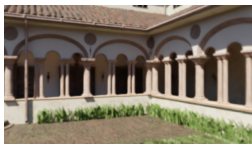
- Neither primitive's normal buffer beats the view-direction control by a usable margin, at equal PSNR.
- The **best reference** is per scene *and* per metric, so no single row can serve as the baseline. Every following table carries both.

Reference runs, qualitative: RGB and depth

QUALITATIVE lone-monk, held-out frame. Depth in turbo over a shared range, near to far.



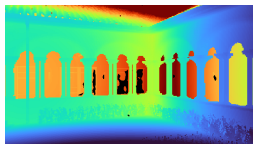
reference RGB



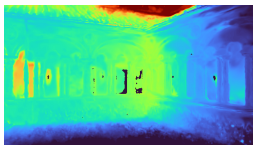
gaussian 36.72 dB



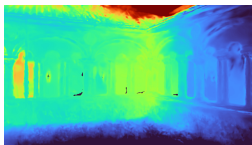
trisurfel 36.71 dB



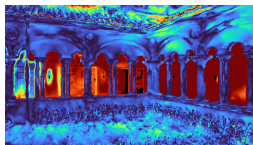
reference depth



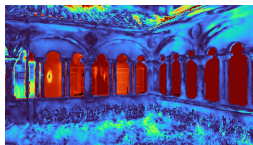
gaussian depth



trisurfel depth



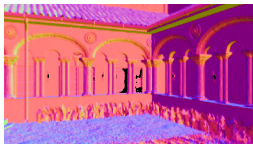
gaussian rel. error



trisurfel rel. error

Reference depth reads the arch openings as **far** (orange). Both references put a surface across them at wall depth (green), and both error maps saturate there. Error maps are capped at the `delta1` boundary, so a saturated pixel is one the metric counts as wrong.

Reference runs, qualitative: normals



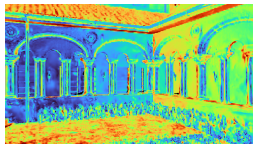
reference normals



gaussian 54.3° , $n_gain -19.1$



trisurfel 35.0° , $n_gain +0.2$



trisurfel angular error

RESULT

- **gaussian**: high-frequency noise, no recoverable surface structure.
- **trisurfel**: smooth and coherent, **but nearly uniform in hue** — most normals point in one direction. That direction is close to the view ray, which is what $n_gain = +0.2$ reports: 19° better than **gaussian** in raw angle, and still no better than the geometry-free control.

CONCLUSION The raw angle and the visual impression both rank **trisurfel** far ahead; the control says neither reference has a normal buffer carrying geometry. Raw angle alone would have declared this solved.

The three scenes fail differently

Scene	floaters px	δ_1 failures	Failure mode
spnza	0.2–2%	moderate	floaters; per-ray spread detects them (AUC 0.68–0.90)
emerald-square	up to 13%	high	floaters, largest depth range
lone-monk	0.01%	17.7%	confidently placed, wrong; no floaters to find

RESULT `lone-monk` has the fewest floaters and the worst δ_1 rate. Its rays are already concentrated, at the wrong distance — visible on the previous slide as an opaque surface across the arch openings.

CONCLUSION A term that works by concentrating a ray cannot help `lone-monk`. Metrics are reported per scene throughout; a scene mean would average this away.

Experiment 1: depth-normal consistency

PREMISE The rendered normal and the rendered depth are produced independently and nothing ties them together. The depth gradient implies a surface orientation; the normal buffer need not agree with it.

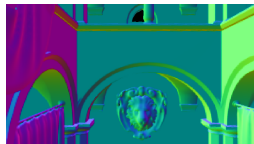
EXPERIMENT Penalise the angle between the rendered normal and the normal implied by the local geometry of the rendered depth. No external data; internal consistency only. Weight 0.05, `depth_normal_from_iter` lowered from 7000 to fit a 7k sweep.

RESULT

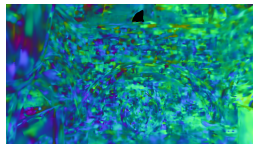
Run	<i>sponza</i>				<i>lone-monk</i>				<i>emerald-square</i>			
	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR
gaussian (ref)	0.0571	52.4	-9.6	36.29	0.0982	54.3	-19.1	36.72	0.1368	65.8	-15.5	32.90
trisurfel (ref)	0.0600	44.7	-1.9	36.19	0.0987	35.0	+0.2	36.71	0.1327	49.4	+0.9	32.92
dn 0.05	0.0474	25.4	+17.4	36.10	0.0976	37.4	-2.1	36.56	0.1309	41.3	+9.0	32.42

Experiment 1: qualitative and conclusion

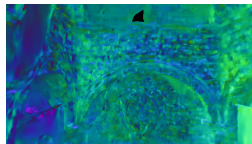
QUALITATIVE `sponza` normals, against the reference and both reference runs.



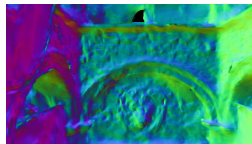
reference



gaussian 52.4°



trisurfel 44.7°



dn 0.05 25.4°

CONCLUSION

- Normals: beats `gaussian` on all three scenes by 17–27°. Against the *best* reference it wins on `sponza` (−19.3°) and `emerald-square` (−8.1°) and **loses on** `lone-monk` (37.4° against `trisurfel`'s 35.0°).
- `n_gain` turns positive on `sponza` and `emerald-square`; on `lone-monk` it stays **negative** (−2.1), as `gaussian`'s does.
- Depth: −17.0% on `sponza`, **under 1%** on `lone-monk` — i.e. nothing on the scene whose depth is worst.
- Cost: 0.16–0.50 dB PSNR.

Landed, default off. The arch, masonry courses and boss are recovered from a buffer that was noise. But `lone-monk` resists it on both metrics, and had the deck compared only against `gaussian` this would have read as a clean win on all three scenes.

Experiment 2: depth variance along the ray

PREMISE A ray that spreads its weight over many distances has no single surface. Penalising the per-ray variance of depth should make each ray commit to one.

EXPERIMENT Render per-ray M_2 , form $\text{Var} = M_2 - D^2/\text{acc}$, penalise it. Then the scale-free form $\text{Var}/\mu^2 = \text{acc} \cdot M_2/D^2 - 1$, which needed no new buffer, kernel or backward pass. Weights 1 and 0.01.

RESULT

Run	sponza				lone-monk				emerald-square			
	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR
gaussian (ref)	0.0571	52.4	-9.6	36.29	0.0982	54.3	-19.1	36.72	0.1368	65.8	-15.5	32.90
trisurfel (ref)	0.0600	44.7	-1.9	36.19	0.0987	35.0	+0.2	36.71	0.1327	49.4	+0.9	32.92
dvrel 1	0.0593	51.8	-8.9	35.27	0.0980	56.1	-20.9	36.65	0.1792	67.6	-17.3	31.13
dvrel 0.01	0.0526	52.4	-9.7	36.33	0.0993	53.2	-18.0	36.80	0.1325	65.7	-15.4	32.87

The absolute form (earlier sweep, not in this table) gave +6% abs_rel on sponza and raised the floater fraction from 0.30% to 1.62%.

From depth_variance_mechanism.py: per-ray spread $\times 0.17$ while wrong | tight rose $\times 170$.

Experiment 2: mechanism and conclusion

WHY In pairwise form,

$$M_2 - \frac{D^2}{\text{acc}} = \frac{1}{2 \text{acc}} \sum_{ij} w_i w_j (t_i - t_j)^2$$

- Variance is zero for a Dirac at *any* distance: the objective says commit, never where.
- Two-hit case is a **double well** in the near opacity, since $w_{\text{near}} w_{\text{far}}$ carries $\alpha(1 - \alpha)$.
- The near well is **absorbing**: at $\alpha = 1$ transmittance zeroes the gradient to everything behind, for every loss. The error is permanent.

CONCLUSION

- **Refuted as posed**: the absolute form did what it was asked — rays six times tighter — and became 170 times more likely to be confidently wrong.
- The relative form at 0.01 improves depth on *sponza* (**−7.9%**) at no PSNR cost, but against the best reference it is **null** on *emerald-square* (−0.2%) and **worse** on *lone-monk* (+1.2%). Normals untouched.
- Identical minima, different behaviour: descent responds to the barrier, not the location of the optima.

RESCALING Dividing by μ^2 leaves both wells at exactly zero, and was first dismissed for that reason.

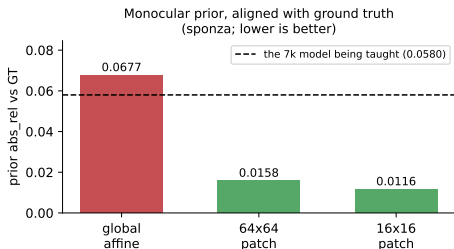
	absolute	relative
separatrix	0.49	0.82
bad basin	51%	18%
sponza	+6%	−7.9%

Experiment 3: regression against a monocular prior

PREMISE A monocular depth model carries information the scene does not. Plan: supervise it with a scale-invariant loss, fitting per-image scale and shift so the predictor's arbitrary affine gauge is not supervised.

EXPERIMENT Before training anything: measure the prior itself. `DepthAnythingV2-Base` on `sponza`, affine-aligned against *ground truth* at decreasing granularity — the most favourable alignment possible, since it uses the reference the loss would not have.

RESULT



- One affine per frame: **0.0677**, against **0.0580** for the 7k model it would be teaching.
- Per 64×64 patch: 0.0158. Per 16×16 : **0.0116**.
- Correlation with $1/z$ is $+0.978$; an affine fit reaches R^2 0.956 in disparity against 0.821 in depth.
- The prior emits z-depth; this renderer's convention is Euclidean ray distance. Radial divergence reaches 20% at the image corners.

CONCLUSION **Abandoned before training.** A globally aligned prior is worse than the model it would teach, and the local numbers say why: the useful structure is not in any single affine gauge, so scale-invariance — the exact remedy planned — does not recover it.

Experiment 4: ordinal supervision from the same prior

PREMISE Only the prior's *ordering* survived the diagnostic. On pixels where the trained model fails `delta1`, the prior is closer to the reference 91% of the time.

EXPERIMENT Read the ordering only. For a pixel pair, penalise the rendered gap only where render and prior disagree about which is nearer:

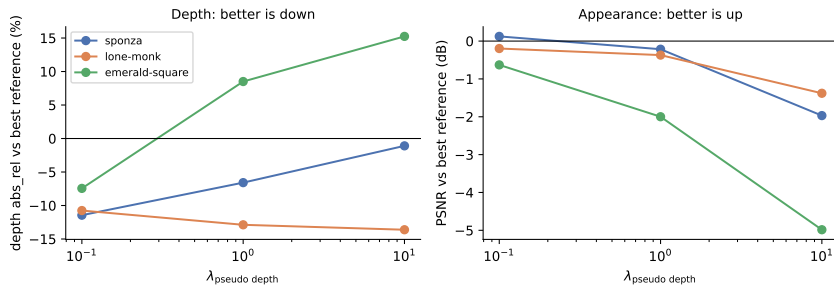
$$\mathcal{L} = \frac{1}{|\mathcal{P}|} \sum_{(a,b) \in \mathcal{P}} \text{relu}(- (d_a - d_b) \cdot \text{sign}(\text{disp}_b - \text{disp}_a)) / \text{scene extent}$$

- Invariant to *any* increasing transform of the prior: disposes of alignment, the disparity inversion and z-vs-ray-distance at once.
- One-sided: a correctly ordered pair contributes zero and no gradient.
- Pairs formed by *cropping* at a random offset $\leq 5\%$ of the image side, not `torch.roll`, which would wrap opposite edges into pairs.

RESULT

Run	sponza				lone-monk				emerald-square			
	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR
gaussian (ref)	0.0571	52.4	-9.6	36.29	0.0982	54.3	-19.1	36.72	0.1368	65.8	-15.5	32.90
trisurfel (ref)	0.0600	44.7	-1.9	36.19	0.0987	35.0	+0.2	36.71	0.1327	49.4	+0.9	32.92
pd 0.1	0.0506	52.5	-9.6	36.41	0.0876	53.1	-17.8	36.52	0.1228	66.2	-15.9	32.29
pd 1	0.0534	52.8	-9.9	36.07	0.0855	53.3	-18.1	36.35	0.1440	69.3	-19.0	30.92
pd 10	0.0565	55.0	-12.2	34.32	0.0848	57.0	-21.8	35.34	0.1529	69.3	-19.0	27.94

Experiment 4: weight sweep



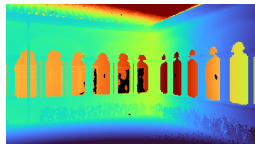
RESULT

- $\lambda = 0.1$: depth -7.4 to -11.5% against the best reference on **all three** scenes, at $>7\sigma$ on each (3-seed means, pooled standard error). First term in this report to improve depth everywhere.
- $\lambda = 1$: more depth on `lone-monk` (-12.9%), but `emerald-square` depth goes **positive** and PSNR costs **2.0 dB**.
- $\lambda = 10$: most of the depth gain given up; $\lambda = 100$ diverges.
- Normals unchanged at every weight — expected, since the rendered normal is the particle's local z axis and depth ordering does not reach it.

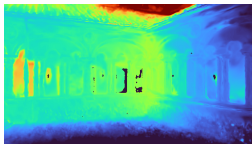
CONCLUSION Usable band is narrow and centred on $\lambda = 0.1$. Cost there: **+0.12 dB** on `sponza`, **-0.20** and **-0.63 dB** on the others.

Experiment 4: qualitative

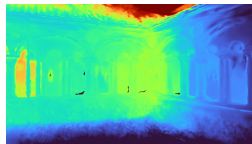
QUALITATIVE lone-monk depth and relative error, against the reference and both reference runs.



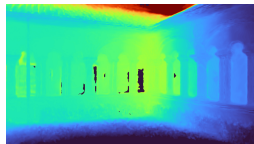
reference depth



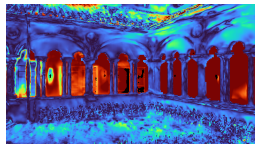
gaussian



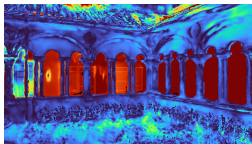
trisurfel



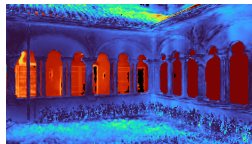
pd 0.1



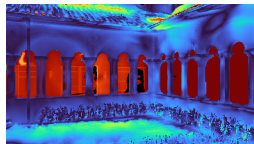
gaussian error



trisurfel error



pd 0.1 error



pd 0.1 + dn error

CONCLUSION

- The arch openings stay **saturated** in every render: the ordinal term does **not** fix them.
- What improves is the diffuse error — roof line, walls, ground all go blue.
- Consistent with the mechanism: pairs are drawn at $\leq 5\%$ offsets and the prior is reliable locally (0.0116 per patch), not globally (0.0677). A hole in a wall is a long-range discontinuity.

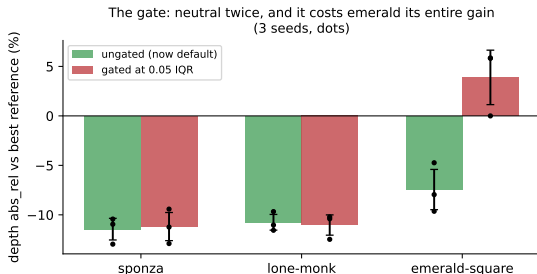
Experiment 5: gating pairs by the prior's confidence

PREMISE Near-tied pairs are where the prior's sign is noise. Offline, on *sponza*:

- ordinal agreement with the reference, all pairs: 84%
- after dropping pairs whose disparity gap is below 0.05 of the map's IQR: **97%**

So ungated, roughly one pair in six pushes the wrong way.

EXPERIMENT Same term, same weight, gate on and off. 3 seeds each.



RESULT

Run	sponza				lone-monk				emerald-square			
	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR
gaussian (ref)	0.0571	52.4	-9.6	36.29	0.0982	54.3	-19.1	36.72	0.1368	65.8	-15.5	32.90
trisurfel (ref)	0.0600	44.7	-1.9	36.19	0.0987	35.0	+0.2	36.71	0.1327	49.4	+0.9	32.92
pd 0.1	0.0506	52.5	-9.6	36.41	0.0876	53.1	-17.8	36.52	0.1228	66.2	-15.9	32.29
pd 0.1 gated	0.0508	52.6	-9.8	36.35	0.0873	54.3	-19.1	36.73	0.1378	67.6	-17.3	32.28

Experiment 5: conclusion

RESULT Neutral on `sponza` and `lone-monk` (within seed noise); on `emerald-square` it removes the entire gain, -7.4% becoming $+3.9\%$.

CONCLUSION **Refuted**. Two reasons, both derivable from numbers already collected before the runs:

- A large disparity gap selects a **long-range** comparison — precisely where this prior is weakest (0.0677 global against 0.0116 per patch). The gate keeps the prior's worst structure and discards its best.
- For a **one-sided** loss, 97% agreement means only 3% of surviving pairs can produce any gradient at all. High agreement is a warning, not a benefit.

The offline metric counted pairs instead of asking what they teach. Default is now off, pinned by a test so re-enabling it must be re-measured.

Experiment 6: combining terms

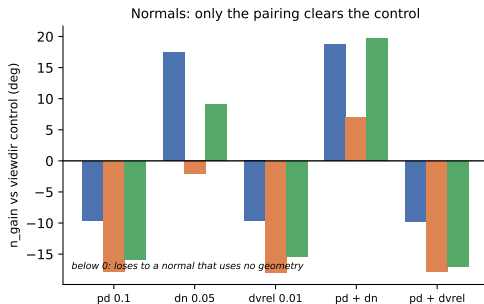
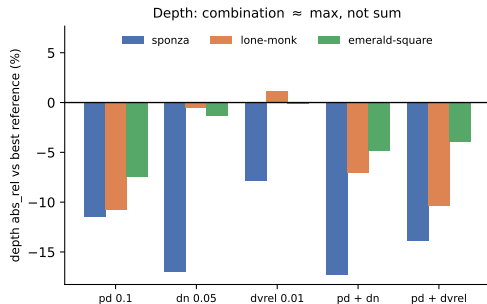
PREMISE The terms target different failures: internal consistency ($_{dn, dvrel}$) against an external prior ($_{pd}$). Per the failure-mode table they should be complementary by scene.

EXPERIMENT Each term at its own measured optimum, and the pairs. Single seed.

RESULT

Run	<i>sponza</i>				<i>lone-monk</i>				<i>emerald-square</i>			
	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR
gaussian (ref)	0.0571	52.4	-9.6	36.29	0.0982	54.3	-19.1	36.72	0.1368	65.8	-15.5	32.90
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pd 0.1	0.0506	52.5	-9.6	36.41	0.0876	53.1	-17.8	36.52	0.1228	66.2	-15.9	32.29
dn 0.05	0.0474	25.4	+17.4	36.10	0.0976	37.4	-2.1	36.56	0.1309	41.3	+9.0	32.42
pd 0.1 + dn 0.05	0.0472	24.0	+18.8	35.77	0.0912	28.3	+6.9	36.50	0.1262	30.5	+19.8	31.97
dvrel 0.01	0.0526	52.4	-9.7	36.33	0.0993	53.2	-18.0	36.80	0.1325	65.7	-15.4	32.87
pd 0.1 + dvrel 0.01	0.0492	52.6	-9.8	36.16	0.0880	53.1	-17.8	36.80	0.1275	67.3	-17.0	32.15

Experiment 6: result

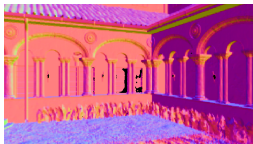


RESULT

- **Depth is sub-additive.** Pairs land near the *max* of their parts, never the sum. On *lone-monk*, *pd + dn* (−7.1%) is **worse than *pd* alone** (−10.8%).
- **Normals are super-additive.** *pd* alone does nothing for normals; *dn* alone leaves *lone-monk* below the control. *pd + dn* is the **only** configuration whose normals clear the control on all three scenes, and it beats *trisurfel* by 6.7–20.7°.
- Mechanism: *dn* ties the normal to the depth *gradient*, so it can only be as good as the depth given to it. The prior improves the depth; consistency carries that into normals.

Experiment 6: qualitative

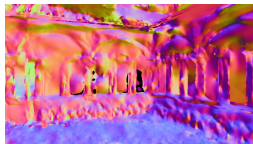
QUALITATIVE lone-monk normals. Same frame, same encoding.



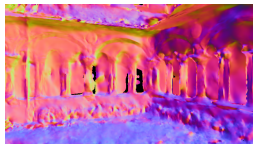
reference



trisurfel 35.0° , $+0.2$



dn 0.05 37.4° , -2.1



pd + dn 28.3° , $+6.9$

CONCLUSION

- dn alone: columns and arches blobby; still **below the control**.
- pd + dn: colonnade resolved, and the only panel here whose n_{gain} is meaningfully positive. Adding a depth term that touches no normals is what flipped the sign.

Experiment 6: the weight trap

EXPERIMENT This pairing was first measured with $\lambda = 1$ for **both** terms — $10\times$ and $100\times$ their measured optima — and written up as a null. **RESULT** `abs_rel`, and the change against each scene's best reference run.

Run	sponza		lone-monk		emerald-square	
gaussian (ref)	0.0571		0.0982		0.1368	
trisurfel (ref)	0.0600		0.0987		0.1327	
pd 0.1	0.0506	−11.4%	0.0876	−10.8%	0.1228	−7.4%
dvrel 0.01	0.0526	−7.9%	0.0993	+1.2%	0.1325	−0.2%
pd 0.1 + dvrel 0.01	0.0492	−13.9%	0.0880	−10.4%	0.1275	−3.9%
pd 1 + dvrel 1	0.0528	−7.6%	0.0855	−12.9%	0.1510	+13.8%

At their own optima the verdict changes from *harmful* to *scene-dependent*: the pair beats both single terms on `sponza`, is a wash on `lone-monk`, and is still worse than `pd` alone on `emerald-square`.

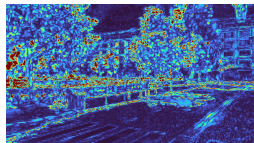
CONCLUSION A null from a combination measured at the wrong weights is not a null about the combination. The over-weighted cell is kept in the sweep so the trap stays visible.

The photometric cost, localised

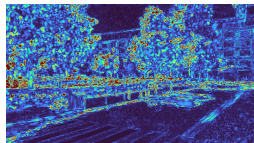
EXPERIMENT PSNR is one number per frame and does not say where a difference sits. Difference every render against the reference image; same colour cap (10% of full range) on every map.



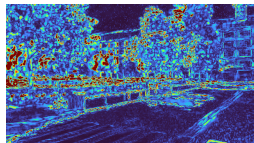
reference RGB



gaussian 32.90 dB



trisurfel 32.92 dB



pd + dn **31.96 dB**

RESULT

- Error in all three is concentrated in **foliage and the hedge line**; the road, buildings and sky are near zero everywhere.
- $pd + dn$ replaces the references' fine speckle with **larger coherent blobs** in the canopy: mean absolute RGB error $4.28 \rightarrow 4.80$ of 255, closer to the reference on only **41%** of pixels.

CONCLUSION The cost is a real regression, not a metric artefact, and it is spatially specific — vegetation, where a consistency term assuming a locally planar surface should be expected to pay most.

The photometric cost, zoomed

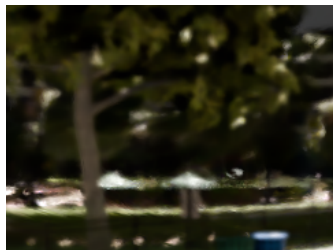
QUALITATIVE Canopy crop of the same frame, against the better reference run here by PSNR. `gaussian` carries the same artifacts and is omitted for space.



reference



trisurfel 32.92 dB



pd + dn 31.97 dB

RESULT Two changes at once:

- The references' **spiky starburst artifacts** (left of frame) are **gone** — they are not in the reference image either.
- Genuine foliage structure is **smoothed away** with them, and against the reference that costs more than the artifacts did.

CONCLUSION Removing visible artifacts and losing PSNR are not contradictory, but here the loss is not paid for: the render moves away from the reference on the majority of pixels.

Summary: depth

RESULT `abs_rel`, and the change against each scene's **best reference run**.

Run	sponza		lone-monk		emerald-square	
gaussian (ref)	0.0571		0.0982		0.1368	
trisurfel (ref)	0.0600		0.0987		0.1327	
pd 0.1	0.0506	−11.4%	0.0876	−10.8%	0.1228	−7.4%
dn 0.05	0.0474	−17.0%	0.0976	−0.5%	0.1309	−1.4%
dvrel 0.01	0.0526	−7.9%	0.0993	+1.2%	0.1325	−0.2%
pd 0.1 + dn 0.05	0.0472	−17.3%	0.0912	−7.1%	0.1262	−4.9%
pd 0.1 + dvrel 0.01	0.0492	−13.9%	0.0880	−10.4%	0.1275	−3.9%

- `pd 0.1` is the only *single* term improving depth on all three scenes by more than seed noise (± 0.002 – 0.004). `dn 0.05`'s `lone-monk` and `emerald-square` gains (0.0005, 0.0018) are inside it; `dvrel` loses on `lone-monk`. Both pairs clear it on all three, at single seed.
- Best per scene: `pd + dn` on `sponza`, `pd` alone on `lone-monk` and `emerald-square`.

Summary: normals

RESULT Mean angle, and the change against each scene's **best reference run** (which is `trisurfel` on all three).

Run	sponza		lone-monk		emerald-square	
gaussian (ref)	52.4		54.3		65.8	
trisurfel (ref)	44.7		35.0		49.4	
pd 0.1	52.5	+17.3%	53.1	+51.7%	66.2	+34.1%
dn 0.05	25.4	-43.1%	37.4	+6.7%	41.3	-16.3%
dvrel 0.01	52.4	+17.3%	53.2	+52.1%	65.7	+33.2%
pd 0.1 + dn 0.05	24.0	-46.3%	28.3	-19.2%	30.5	-38.2%
pd 0.1 + dvrel 0.01	52.6	+17.7%	53.1	+51.6%	67.3	+36.4%

- `pd` and `dvrel` leave normals unchanged; both act on depth only.
- `pd + dn` is best on all three scenes, and the only configuration with positive `n_gain` everywhere.

Status and settings

Term	Status	Verdict
Depth–normal consistency	landed, off	Recovers normals; lone-monk still loses to trisurfel
Depth variance, absolute	refuted	Confidence without accuracy; wrong tight $\times 170$
Depth variance, relative	landed, off	Depth on sponza only; null elsewhere, no cost
Scale-z (PGSR port)	landed, small	Only after fixing the normal-axis assumption
Pseudo-depth regression	abandoned	Prior worse than the model it would teach
Pseudo-depth ordinal	landed, off	Only single term improving depth on all 3 scenes
+ confidence gate	refuted	Keeps the prior's worst structure
Multi-view consistency, mesh	not started	

- Depth, one term: $pd\ 0.1$. -7 to -11% against the best reference on all three scenes, $+0.12$ to -0.63 dB.
- Depth and normals: $pd\ 0.1 + dn\ 0.05$. Best normals everywhere, -0.22 to -0.95 dB.
- All remain **off by default**: the photometric cost is real and PSNR is what the codebase is currently evaluated on.

Limitations

- **7k iterations only.** No term confirmed at 30k. Iteration-gated defaults (`*_from_iter`) were lowered for the sweep and do not transfer unchanged.
- **Three scenes**, chosen because they fail differently, not because they are representative.
- **Combination cells are single-seed** against 3-seed references. The `lone-monk` `n_gain` sign flip and the `sponza` depth gains are far outside seed noise; `emerald-square` differences of ~ 1 point are not.
- **Why `emerald-square` pays most is unexplained.** That the cost is concentrated in foliage is measured; the link to its depth range is a hypothesis.
- **No mesh.** Nothing here is evaluated by surface extraction or Chamfer distance.
- **The prior diagnostic is `sponza`-only**, and `DepthAnythingV2-Base` only.

Method notes that changed a conclusion

1. **Include a control that uses no information.** `gaussian`'s normal buffer looked reasonable and lost to pointing every normal at the camera; `trisurfel`'s looks good, scores 19° better, and still only ties the control.
2. **Report per scene.** `lone-monk` has no floaters and the worst depth; a whole family of terms cannot help it, and a scene mean reported a win.
3. **Compare against the strongest reference, not the one you modified.** Every treatment here is a change to `gaussian`, so `gaussian` was the natural baseline — but `trisurfel` is better on normals everywhere and on `emerald-square` depth. Re-scoring against the best of the two removed a claim that `dn 0.05` beat all references on normals, and cut `pd 0.1`'s `emerald-square` depth gain from 10.3% to 7.4%.
4. **Measure the reference quantity before building on it.** The prior diagnostic cancelled a planned loss; the PGSR port regressed normals 30° because “min scale” is not this renderer's normal axis.
5. **An offline metric can motivate the wrong mechanism.** The gate's 84%→97% was real and irrelevant.
6. **Measure combinations at each term's own optimum**, and against each single term, not only against a baseline.
7. **Localise a metric before interpreting it.** The RGB difference maps turned an assumed metric artefact into a measured regression.